

Benefits and ROI—A customer's feedback

ELICO Ltd deployed Crux LX last year in all three offices in India—Hyderabad, Chennai and Mysore. Ramesh Datla, managing director, ELICO Ltd and ex-chairman, Confederation of Indian Industry (CII), says, "We were using Siemens PABX for more than 15 years. Over the years, we expanded and there is an increasing demand for more connections. The major challenge with the existing PABX was expandability and serviceability because of age. And, upgrade was hindered because of cost of other new solutions."

"Deploying Crux LX is beneficial because capital cost and setup time are much lower and maintenance cost is negligible. We use Google Mesh wireless infrastructure, and integrating Crux LX to it was quite straightforward. Since everyone uses cellphones these days, it is accessible to all, and cross-location communication cost is greatly reduced. Expansion of the solution is also easy."

About the payback period, Datla mentions, "The ROI is immediate. A traditional PABX for 50 or 100 lines costs a lot of money. With Crux solution, both can coexist. There are no additional costs involved after installation."

can route call data through the router and land directly to the smartphone instead—all as an on-premise solution instead of a cloud-based one.

"Our product is basically a wireless router that performs telephony and IP-PBX operations. The idea is to have a router that can utilise and analyse all data without having to wait for it to reach the cloud or depend on the Internet," says Gupta.

Crux LX is designed based on open source resources. Software setup is Linux Apache MySQL PHP (LAMP) architecture with MQTT messaging broker and built-in support for Node.js. For hardware, it uses Raspberry Pi as the base board and Broadcom processor. Avnet and Element14 are the supply partners for Crux Labs.

The platform can be operated using any compatible open source mobile application (like Zoiper and GSWave) available in Apple's App Store and Google's Play Store. It features voice calls, conference bridges, voicemail and many other useful options.

One Crux LX unit can enroll up to 250 users and support 20 concurrent conversations. Users can pair multiple Crux devices, even across geographies over the Internet, to expand user support and create an internal network.

Gupta adds, "We are in contact with some of our partners who have their applications on a cloud service (like Amazon Web Services) and are exploring to bring those applications on top of Crux LX platform. This will create a full-scale on-premise appliance for them without any dependence and expenditure on cloud, or any additional setup."

Overcoming technical barriers

Picking up pace in telecommunications science and finding a solution unaffected by today's digital clutter were the first set of challenges Gupta had to face. Initially, the solution was planned to be manufactured in India, with the team following up some factories and gathering estimates. He says, "Cost points quoted by factories were prohibitory, costing almost ₹ 20 to 30 million for the production process, which had to be paid upfront." Taking an open systems approach, the team was able to manufacture the product at a much lower cost.

Team Crux overcame most challenges with the help of Board of Advisors, which comprised family members and friends who were also experts in different fields of operation, including technology, business strategy, marketing and external communications.

Crux timeline

The idea of Crux LX was conceived in October 2016. Towards the end of December 2016, the team had built a coverless prototype after consultation with

advisors and some small businesses. March 2017 marked the start of Alpha launch, with one of the team's clients based out of Singapore.

The following April, the first official unit of Crux LX was brought to life. The period from July 2017 to September 2017 marked the product's Beta phase, where the solution was deployed to the first Indian client, ELICO Ltd, a Hyderabad-based analytical instruments manufacturer.

In March 2018, Crux LX was registered with Info-communications Media Development Authority (IMDA) Singapore. It has been trading in the open market since, and has achieved authorised vendor status for Singtel, Singapore.

India connect and roadmap ahead

Crux Labs is set to open an office in Bengaluru, India in another six to eight months. With expansion plans, the company is fine-tuning its business operations by involving digital technology in areas like CRM and accounting, among others. The aim is to create a strong foothold among the Indian SMB chain.

However, Gupta foresees a couple of challenges. First, especially in tier 2 and tier 3 cities, lack of proper smartphone and basic resources with a large number of users may hinder the user experience. Therefore the company will start its services with tier 1 cities initially.

The second challenge lies in the lack of proper supplier and distributor chain for SMBs, like those available for big enterprises. The strategy is to deliver the products initially and slowly create a distributor chain.

In the long run, Crux Labs plans to reach out to the consumer and domestic segment as well. For that, it will scale down Crux LX into a more compact and plug-and-play form factor at cheaper cost. It will also venture out and expand into other industries to provide solutions across different verticals. For instance, the team is working on call accounting technologies to enable per user billing and call analytics, which is useful for hotels and other commercial entities. It is also looking into VSAT voice communication that can be used for maritime communication, or in remote areas.



How Crux LX works